

PC2174A: Mathematical Methods in Physics I

Assignment 3

1. [20 pts] If $I_n = \int_0^{\pi/2} e^{ax} \sin^n x \, dx$, where a is a constant, show that

$$I_n = \frac{a e^{a\pi/2}}{n^2 + a^2} + \frac{n(n-1)}{n^2 + a^2} I_{n-2}, \quad n \geq 2.$$

Answer:

It is obvious that

$$\begin{aligned} I_n &= \int_0^{\pi/2} e^{ax} \sin^n x \, dx = \frac{1}{a} \sin^n x e^{ax} \Big|_0^{\pi/2} - \frac{1}{a} \int_0^{\pi/2} n \sin^{n-1} x \cos x e^{ax} \, dx \\ &= \frac{e^{a\pi/2}}{a} - \frac{n}{a^2} \int_0^{\pi/2} \sin^{n-1} x \cos x \, d(e^{ax}) \\ &= \frac{e^{a\pi/2}}{a} - \frac{n}{a^2} \sin^{n-1} x \cos x e^{ax} \Big|_0^{\pi/2} + \frac{n}{a^2} \int_0^{\pi/2} e^{ax} \, d(\sin^{n-1} x \cos x) \\ &= \frac{e^{a\pi/2}}{a} + \frac{n}{a^2} \int_0^{\pi/2} e^{ax} ((n-1) \sin^{n-2} x \cos^2 x - \sin^n x) \, dx \\ &= \frac{e^{a\pi/2}}{a} + \frac{n(n-1)}{a^2} \int_0^{\pi/2} e^{ax} \sin^{n-2} x (1 - \sin^2 x) \, dx - \frac{n}{a^2} \int_0^{\pi/2} e^{ax} \sin^n x \, dx \\ &= \frac{e^{a\pi/2}}{a} + \frac{n(n-1)}{a^2} I_{n-2} - \frac{n(n-1)}{a^2} I_n - \frac{n}{a^2} I_n \end{aligned} \tag{1}$$

namely

$$I_n = \frac{a e^{a\pi/2}}{n^2 + a^2} + \frac{n(n-1)}{n^2 + a^2} I_{n-2}, \quad n \geq 2.$$

Q.E.D.

2. [30 pts] An important class of integrals is as follows: $\int F(\sin \theta, \cos \theta) \, d\theta$, where $F(\sin \theta, \cos \theta)$ is a rational function of $\sin \theta$ and $\cos \theta$. One possible way to evaluate the above integral is known as the *tangent half-angle substitution*: $t \equiv \tan \frac{\theta}{2}$, which converts it into an integral involving a rational function of t . Evaluate the following integrals:

(a) $\int \frac{1}{2 \sin x - \cos x + 5} \, dx$

(b) $\int \frac{1}{4 + 5 \cos(x/2)} \, dx$

(c) $\int \frac{1}{1 + \csc x} \, dx$

(d) $\int \frac{1}{\sin^4 x + \cos^4 x} \, dx$

Answer:

(a)

$$\begin{aligned} OE &= \int \frac{\sin^2 \frac{x}{2} + \cos^2 \frac{x}{2}}{4 \sin \frac{x}{2} \cos \frac{x}{2} - 2 \cos^2 \frac{x}{2} + 6 \sin^2 \frac{x}{2} + 6 \cos^2 \frac{x}{2}} \, dx \\ &= \int \frac{1 + \tan^2 \frac{x}{2}}{4 \tan \frac{x}{2} - 2 + 6 \tan^2 \frac{x}{2} + 6} \, dx = \int \frac{dt}{3t^2 + 2t + 2} \quad (t = \tan \frac{x}{2}) \\ &= \frac{1}{\sqrt{5}} \arctan \left[\frac{3}{\sqrt{5}} \left(\tan \frac{x}{2} + \frac{1}{3} \right) \right] + C \end{aligned} \tag{2}$$

(b)

$$\begin{aligned}
 OE &= \int \frac{dx}{4 + 5 \cos(x/2)} = \int \frac{\sin^2 \frac{x}{4} + \cos^2 \frac{x}{4}}{4 \sin^2 \frac{x}{4} + 4 \cos^2 \frac{x}{4} + 5 \cos^2 \frac{x}{4} - 5 \sin^2 \frac{x}{4}} dx \\
 &= \int \frac{\tan^2 \frac{x}{4} + 1}{9 - \tan^2 \frac{x}{4}} dx = \int \frac{4}{9 - t^2} dt \quad (t = \tan \frac{x}{4}) = \frac{2}{3} \ln \left| \frac{3 + \tan \frac{x}{4}}{3 - \tan \frac{x}{4}} \right| + C
 \end{aligned} \tag{3}$$

(c)

$$\begin{aligned}
 OE &= \int \frac{2 \sin \frac{x}{2} \cos \frac{x}{2} dx}{2 \sin \frac{x}{2} \cos \frac{x}{2} + \sin^2 \frac{x}{2} + \cos^2 \frac{x}{2}} \\
 &= \int \frac{2 \tan \frac{x}{2} dx}{2 \tan \frac{x}{2} + 1 + \tan^2 \frac{x}{2}} = \int \frac{4t}{(t+1)^2(t^2+1)} dt \quad (t = \tan \frac{x}{2}) \\
 &= \int \left[-\frac{2}{(t+1)^2} + \frac{2}{t^2+1} \right] dt = \frac{2}{\tan \frac{x}{2} + 1} + x + C
 \end{aligned} \tag{4}$$

(d)

$$\begin{aligned}
 OE &= \int \frac{\sin^4 x + 2 \sin^2 x \cos^2 x + \cos^4 x}{\sin^4 x + \cos^4 x} dx \\
 &= \int \frac{\tan^4 x + 2 \tan^2 x + 1}{\tan^4 x + 1} dx = \int \frac{t^2 + 1}{t^4 + 1} dt \quad (t = \tan x) \\
 &= \frac{1}{2} \int \frac{dt}{t^2 + \sqrt{2}t + 1} + \frac{1}{2} \int \frac{dt}{t^2 - \sqrt{2}t + 1} \\
 &= \sqrt{2} \left[\arctan(\sqrt{2} \tan x + 1) + \arctan(\sqrt{2} \tan x - 1) \right] + C
 \end{aligned} \tag{5}$$

3. [25 pts] The gamma function $\Gamma(x)$ is defined by

$$\Gamma(x) \equiv \int_0^\infty \xi^{x-1} e^{-\xi} d\xi, \quad x > 0.$$

(a) Establish a recurrence relationship between $\Gamma(x+1)$ and $\Gamma(x)$. Hence, deduce (i) the value of $\Gamma(x+1)$ when x is a non-negative integer, and (ii) the value of $\Gamma(\frac{7}{2})$ given that $\Gamma(\frac{1}{2}) = \sqrt{\pi}$.

(b) Now, factorial function for any m can be defined by $m! \equiv \Gamma(m+1)$. Evaluate $(-\frac{7}{2})!$.

Answer:

(a) By integration by part,

$$\Gamma(x+1) = \int_0^\infty \xi^x e^{-\xi} d\xi = - \int_0^\infty \xi^x d(e^{-\xi}) = x \int_0^\infty \xi^{x-1} e^{-\xi} d\xi = x\Gamma(x) \tag{6}$$

hence, the relationship of $\Gamma(\frac{7}{2})$ and $\Gamma(\frac{1}{2}) = \sqrt{\pi}$ is:

$$\Gamma(\frac{7}{2}) = \frac{5}{2} \cdot \frac{3}{2} \cdot \frac{1}{2} \cdot \Gamma(\frac{1}{2}) = \frac{15\sqrt{\pi}}{8} \tag{7}$$

(b)

$$(-\frac{7}{2})! = \Gamma(-\frac{5}{2}) = \left(-\frac{2}{5}\right) \left(-\frac{2}{3}\right) \left(-\frac{2}{1}\right) \Gamma\left(\frac{1}{2}\right) = -\frac{8\sqrt{\pi}}{15} \tag{8}$$

4. [25 pts] A rocket fired vertically into the air at $t = 0$ has a profile of acceleration

$$a(t) = \frac{dv(t)}{dt} = \frac{2g}{(1-\alpha t)} - g,$$

for five seconds, where $\alpha = 0.1 \text{ s}^{-1}$ and $g = 9.8 \text{ m s}^{-2}$ is the acceleration due to gravity. Then the fuel runs out and it subsequently moves freely under gravity. What maximum height does the rocket reach?

Answer:

when $0 \leq t \leq 5 \text{ s}$,

$$a(t) = \frac{2g}{(1-\alpha t)} - g \tag{9}$$

then,

$$v(t) = \int_0^t a(t') dt' = \int_0^t \left(\frac{2g}{(1-\alpha t)} - g \right) dt' = -\frac{2g}{\alpha} \ln(1-\alpha t) - gt \quad (10)$$

when fuel is run out, $t_1 = 5$ s

$$v_1 = v(5) = 86.857 \text{ m/s} \quad (11)$$

the location is:

$$\begin{aligned} x(t) &= \int_0^t v(t') dt' = \int_0^t \left(-\frac{2g}{\alpha} \ln(1-\alpha t) - gt \right) dt' \\ &= \frac{2g}{\alpha^2} (1-\alpha t) \ln(1-\alpha t) - \frac{2g}{\alpha^2} (1-\alpha t) + \frac{2g}{\alpha^2} - \frac{1}{2} gt^2 \end{aligned} \quad (12)$$

The height when fuel is run out:

$$x_1 = 178.216 \text{ m} \quad (13)$$

After $t > 5$, $a = -g$

$$v(u) = v_1 - gu, \quad t_{\max} = \frac{v_1}{g} = 5.259 \text{ s} \quad (14)$$

The height it flies after fuel is run out

$$\Delta x = \frac{v_1^2}{2g} = 384.905 \text{ m} \quad (15)$$

So the total is:

$$H_{\max} = 563.121 \text{ m} \quad (16)$$

5. [25 pts] An ellipse centered at the origin in the xy -plane is given by

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1, \quad a > b.$$

(a) Find the area enclosed by the ellipse.

(b) A complete elliptic integral of the second kind is defined as $E(k) \equiv \int_0^{\pi/2} \sqrt{1 - k^2 \sin^2 \theta} d\theta$, $0 \leq k \leq 1$. Find the perimeter of the above ellipse in terms of $E(k)$.

(c) A prolate spheroid is obtained by rotating the above ellipse about the x -axis. Find its volume and surface area.

Answer:

(a) The curve above the x -axis is

$$y = b \sqrt{1 - \frac{x^2}{a^2}}, \quad x \in [-a, a] \quad (17)$$

As a result, the total area enclosed by the ellipse is

$$S = 2 \int_{-a}^a dx \, b \sqrt{1 - \frac{x^2}{a^2}} = 2 \int_{-a}^a d(a \sin \theta) \, b \cos \theta = \pi ab \quad (18)$$

(b) The equivalent parameter representation is $x = a \cos \theta$ and $y = b \sin \theta$, and it is trivial to calculate the perimeter with the symmetry:

$$\begin{aligned} l &= \oint \sqrt{dx^2 + dy^2} = 4 \int_0^{\pi/2} d\theta \sqrt{a^2 \sin^2 \theta + b^2 \cos^2 \theta} = 4 \int_0^{\pi/2} d\theta \sqrt{b^2 \sin^2 \theta + a^2 \cos^2 \theta} \\ &= 4a \int_0^{\pi/2} \sqrt{1 - \frac{a^2 - b^2}{a^2} \sin^2 \theta} d\theta = 4aE\left(\sqrt{\frac{a^2 - b^2}{a^2}}\right) \end{aligned} \quad (19)$$

(c)

$$dV = \pi y^2 dx, \quad V = \int_{-a}^a dx \, \pi b^2 \left(1 - \frac{x^2}{a^2}\right) = \frac{4}{3} \pi ab^2 \quad (20)$$

The surface of the solid is

$$\begin{aligned}
 S &= \int_{half} 2\pi|x| \, dl = 2\pi \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} d\theta \sqrt{a^2 \sin^2 \theta + b^2 \cos^2 \theta} \, a \cos \theta \\
 &= 2\pi a \int_{-1}^1 dx \sqrt{(a^2 - b^2)x^2 + b^2} \\
 &= 2\pi a \left[a + \frac{a^2 - b^2}{b^2} \sinh^{-1} \left(\frac{b}{\sqrt{a^2 - b^2}} \right) \right]
 \end{aligned} \tag{21}$$

or equivalently it has another form:

$$S = 2\pi b^2 \left(1 + \frac{a}{be} \arcsin e \right), \quad e = \sqrt{1 - \frac{a^2}{b^2}} \tag{22}$$

6. [30 pts] The equation of a cardioid in polar coordinates is $\rho = a(1 - \sin \phi)$, $0 \leq \phi < 2\pi$. Find

- (a) its area,
- (b) its total length,
- (c) the surface area of the solid formed by rotating the cardioid about its axis of symmetry, and
- (d) the volume of the same solid.

Answer:

(a)

$$S = \oint \frac{1}{2} \rho^2 d\theta = \frac{1}{2} \int_0^{2\pi} a^2 (1 - \sin \phi)^2 d\phi = \frac{3}{2} \pi a^2 \tag{23}$$

(b)

$$\begin{aligned}
 l &= \oint d\phi \sqrt{\left(\frac{d\rho}{d\phi} \right)^2 + \rho^2} \\
 &= \int_0^{2\pi} d\phi \sqrt{(a \cos \phi)^2 + a^2 (1 - \sin \phi)^2} = \sqrt{2} a \int_0^{2\pi} d\phi |\sin \phi - \cos \phi| \\
 &= \sqrt{2} a \left[\left(\int_0^{\frac{\pi}{4}} + \int_{\frac{5\pi}{4}}^{2\pi} \right) (\cos \phi - \sin \phi) d\phi + \int_{\frac{\pi}{4}}^{\frac{5\pi}{4}} (\sin \phi - \cos \phi) d\phi \right] \\
 &= 8a
 \end{aligned} \tag{24}$$

(c) The surface of rotated cardioid can be calculated by:

$$S = 2\pi \int_{half} |x| \, dl \tag{25}$$

and

$$x = \rho \cos \phi = a \cos \phi (1 - \sin \phi), \quad \phi \in [0, \pi] \tag{26}$$

As a result,

$$S = 2\pi \left(\int_0^{\frac{\pi}{2}} + \int_{\frac{3\pi}{2}}^{2\pi} \right) d\phi \sqrt{2} a \sqrt{1 - \sin \phi} a \cos \phi (1 - \sin \phi) = \frac{32}{5} \pi a^2 \tag{27}$$

(d) We first derive functions and differential in Cardinal coordinate:

$$\begin{aligned}
 x &= \rho \cos \phi = a \cos \phi (1 - \sin \phi), \quad dx = (-a \sin \phi + a \sin^2 \phi - a \cos^2 \phi) d\phi \\
 y &= \rho \sin \phi = a \sin \phi (1 - \sin \phi), \quad dy = (a \cos \phi - 2a \sin \phi \cos \phi) d\phi
 \end{aligned} \tag{28}$$

The volume can be calculated by dividing it into two parts, above the x-axis and below the x-axis. The part above the x-axis, V_1 , can be easily work out by shell method

$$\begin{aligned}
 V_1 &= \int_0^a dx \, 2xy\pi = 2\pi \int_{\frac{\pi}{2}}^0 d\phi \, a(-\sin \phi + 2\sin^2 \phi - 1) a^2 \cos \phi (1 - \sin \phi)^2 \sin \phi \\
 &= 2\pi a^3 \int_1^0 dt \, (2t^2 - t - 1)(1 - t)^2 t = \frac{1}{6} \pi a^3
 \end{aligned} \tag{29}$$

The part below x-axis can be calculated by disk method:

$$\begin{aligned} V_2 &= \int_{-2a}^0 dy \pi x^2 = \pi \int_{\frac{3\pi}{2}}^{2\pi} d\phi a(\cos \phi - 2 \sin \phi \cos \phi) a^2 \cos^2 \phi (1 - \sin \phi)^2 \\ &= \pi a^3 \int_{-1}^0 dt (1 - 2t)(1 - t^2)(1 - t)^2 = \frac{5}{2} \pi a^3 \end{aligned} \quad (30)$$

As the result, the total volume of the solid is

$$V = V_1 + V_2 = \frac{8}{3} \pi a^3 \quad (31)$$

7. [25 pts] A non-uniform rod of mass M lies on the x -axis from $x = -L/2$ to $x = +L/2$. The linear mass density of the rod is proportional to the distance from its lighter end, i.e. $x = -L/2$.

(a) The concept of center of mass plays an important role in translational dynamics. Given a system of point masses $m_1, m_2, \dots, m_N$ located at $x_1, x_2, \dots, x_N$ respectively on the x -axis, the center of mass of the system is defined by

$$x_{\text{CM}} \equiv \frac{1}{M} \sum_{i=1}^N m_i x_i, \quad M \equiv \sum_{i=1}^N m_i,$$

where M is the total mass of the system. Find the center of mass of the non-uniform rod.

(b) The concept of moment of inertia plays an important role in rotational dynamics. Given a system of point masses $m_1, m_2, \dots, m_N$, the moment of inertia of the system about a rotational axis is defined by

$$I \equiv \sum_{i=1}^N m_i r_i^2,$$

where r_i is the perpendicular distance from the point mass m_i to the axis of rotation. Find the moment of inertia of the non-uniform rod about a rotational axis perpendicular to the rod and passing through its (i) heavier end; (ii) lighter end; and (iii) center of mass, respectively.

Answer:

(a) The mass distribution can be written as

$$\lambda(x) = \frac{2M}{L^2} \left(x + \frac{L}{2}\right) \quad (32)$$

$$x_{\text{CM}} = \frac{1}{M} \int_{-\frac{L}{2}}^{\frac{L}{2}} dx x \frac{2M}{L^2} \left(x + \frac{L}{2}\right) = \frac{L}{6} \quad (33)$$

(b) Every small pieces' moment of inertia is

$$dI = dm m^2(r) = dx \frac{2M}{L^2} \left(x + \frac{L}{2}\right) r^2(x) \quad (34)$$

(b.i) When rotating about its heavier end, $r(x) = (\frac{L}{2} - x)$, the moment of inertia is

$$I = \frac{2M}{L^2} \int_{-\frac{L}{2}}^{\frac{L}{2}} dx \left(x + \frac{L}{2}\right) \left(\frac{L}{2} - x\right)^2 = \frac{ML^2}{6} \quad (35)$$

(b.ii) When rotating about its lighter end, $r(x) = (-\frac{L}{2} - x)$, the moment of inertia is

$$I = \frac{2M}{L^2} \int_{-\frac{L}{2}}^{\frac{L}{2}} dx \left(x + \frac{L}{2}\right) \left(-\frac{L}{2} - x\right)^2 = \frac{ML^2}{2} \quad (36)$$

(b.iii) When rotating about its center of mass, $r(x) = (\frac{L}{6} - x)$, the moment of inertia is

$$I = \frac{2M}{L^2} \int_{-\frac{L}{2}}^{\frac{L}{2}} dx \left(x + \frac{L}{2}\right) \left(\frac{L}{6} - x\right)^2 = \frac{ML^2}{18} \quad (37)$$

8. [20 pts] One of the mechanisms of heat transfer is *conduction* which occurs within an object or between two objects in contact. In conduction, the direction of heat flow is always from higher to lower temperature. Consider a rod of conducting material with cross-sectional area A and length L as shown. The left end of the rod is kept at a temperature T_H and the right end at a lower temperature T_C . Experiments show that the rate of heat flow, dQ/dt , is proportional to the cross-sectional area A of the rod and to the temperature difference $T_H - T_C$, and is inversely proportional to the rod length L :

$$\frac{dQ}{dt} = kA \frac{T_H - T_C}{L},$$

where k is the thermal conductivity of the material.

- (a) Two slabs of cross-sectional area A and of thickness L_1 and L_2 and thermal conductivities k_1 and k_2 are arranged in contact face to face. The outer face of the first slab is maintained at temperature T_H and that of the second one at T_C . Find the equivalent conductivity.
- (b) The inside of a hollow cylinder is maintained at a temperature T_H and the outside is at a lower temperature T_C . The wall of the cylinder has thermal conductivity k . The inner and outer radii of the hollow cylinder are a and b respectively, and the length of the hollow cylinder is L . Ignoring end effects, find the rate of heat flow from the inner surface to the outer surface in the radial direction under steady-state conditions at which the rate of heat flow through the material is the same everywhere.

Answer:

- (a) The system in thermal equilibrium is analogous to a steady-state electrical circuit obeying Ohm's law, where the temperature is analogous to the voltage and the rate of heat flow is analogous to the current. So we can define the "thermal resistance" as

$$R_H = \frac{L}{kA} \quad (38)$$

Considering the electric resistance $R = \frac{\rho L}{A}$, thermal resistance has similar form to it, hence obeying the same series and parallel rules. In this question, two slabs are in series, so the total resistance is

$$R_H = \frac{L_1 + L_2}{kA} = \frac{L_1}{k_1 A} + \frac{L_2}{k_2 A} \quad (39)$$

As a result, the equivalent conductivity is

$$k = \frac{L_1 + L_2}{\frac{L_1}{k_1} + \frac{L_2}{k_2}} \quad (40)$$

- (b) Every small pieces of "heat resistances" is

$$dR_H = \frac{dr}{2\pi r L k} \quad (41)$$

The total heat resistance is derived by integrated from a to b

$$R_H = \int_a^b \frac{dr}{2\pi r L k} = \frac{1}{2\pi L k} \ln\left(\frac{b}{a}\right) \quad (42)$$

As a result, the rate of heat flow is

$$\frac{dQ}{dt} = \frac{2\pi L k}{\ln\left(\frac{b}{a}\right)} (T_H - T_C) \quad (43)$$